MotionSmith: A Sketch-Based Design System for Automata Making

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PDF · DOI · Code · Project · Video

Abstract

We present MotionSmith, a sketch-based computational design system developed through participatory design (PD) with three experienced makers of mechanical crafts (automata). MotionSmith enables users to sketch and iteratively identify a desired motion, explore system-generated mechanisms to realize the motion, refine the chosen mechanism, and export fabrication-ready files. We developed the system through a year-long iterative process, including PD with three artists. Insights from this collaboration emphasized the importance of prioritizing creative intent over literal sketch fidelity, enabling fluent iteration across design stages, and ensuring mechanically simple, fabricable outcomes. The resulting system leverages computational support while maintaining user agency across sketching, mechanism synthesis, and blueprint generation. Our deployment with experts shows how this workflow bridges user intent and mechanical realization and surfaces opportunities to expand educational scaffolding and fabrication guidance for a broader community of makers.

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